Jocelyn Zhao

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Education

Massachusetts Institute of Technology Expected May 2027
*Candidate for Bachelor of Science in **Computer Science and Engineering** (GPA: 4.9 / 5.0)* *Cambridge, MA*

- **Relevant Coursework:** Software Construction, Design and Analysis of Algorithms, Computational Structures, Machine Learning, Deep Learning, Natural Language Processing, Robotic Manipulation
- **Activities:** HKN, Solar Electric Vehicle Team

Work Experience

Stripe May 2026 – Aug 2026
Software Engineering Intern *Seattle, WA*

- Designed, built, and shipped a Checkout Sessions feature for express payments and shipping events using TypeScript and Ruby
- Partnered with senior engineers to design concurrency-safe event handling for asynchronous checkout workflows
- Added production observability with event metrics, error categories, and LogScale/Hubble Query dashboards

MIT CSAIL (CLEAR Lab) Mar 2026 – May 2026
Undergraduate Researcher *Cambridge, MA*

- Implemented NVIDIA's geometric fabrics controller for the Franka Emika robot arm
- Reduced control-loop latency from 150 ms to 16 ms by minimizing remote procedure calls (RPCs)

MIT EECS Feb 2026 – May 2026
Teaching Staff for Intro to Machine Learning *Cambridge, MA*

- Spent 3.5 hours a week explaining machine learning concepts to students, collaborated with 54 instructors and assistants to answer student questions and adjust course materials

ReHome Jun 2025 – Aug 2025
Software Engineer Intern *Cambridge, MA*

- Built automated data migration tool, reducing manual column mapping by 60% using Hugging Face transformers
- Developed full-stack Django features backed by PostgreSQL with Agile practices, writing unit and integration tests
- Collaborated with senior engineer on building Alpine + uv Docker image and setting up AWS Lightsail instance

Youngstown State University Jun 2022 – Aug 2022
Data Science Intern *Youngstown, OH*

- Developed and trained deep neural networks in PyTorch for clothing image classification, optimized model performance through hyperparameter tuning to achieve over 90% accuracy

Technical Projects

LLM Alignment | *PyTorch, Hugging Face, LoRA*

- Improved the safety of TinyLlama-1.1B via LoRA fine-tuning, achieving robustness to adversarial prompts on AdvBench and JailbreakBench
- Conducted benchmarking and ablation experiments, showing improved performance tradeoffs over DPO

BattleLingo | *MongoDB, Express.js, React, Node.js*

- Secured 4th place out of 60 submissions in a competitive web app showcase with a real-time multiplayer typing game designed to support language learning, built as a team of 3
- Designed and launched a live PVP/bot typing game leveraging Socket.io to handle real-time communication and concurrency, supporting nearly 100 users with responsive gameplay

SegFormer Reimplementation for Image Segmentation | *Tensorflow, Keras, Numpy*

- Built a lightweight image segmentation model based on published paper, consists of 3.7M trainable parameters
- Achieved around 70% mean Intersection over Union (mIoU) on KITTI road detection dataset

Skills

Languages: Python, JavaScript, C, Typescript, Ruby, HTML, Tailwind CSS
Technologies: PyTorch, Tensorflow/Keras, Django, Flask, Docker
Version Control: git, GitHub, GitLab